

AnthracoForm

User guide — AnthracoForm v1.3 (CSV schema v1.3)

Supplementary material S3. Software: <https://doi.org/10.5281/zenodo.20487673> (MIT licence). Data dictionary: S1. Example analyses in R: S2.

This guide explains how to record wood-charcoal fragments with AnthracoForm, step by step. The screenshots show the Saharo-Sindian (Sindh) module with a demonstration file (site code DEMO); the Eastern Mediterranean (Lebanon) module works in the same way and differs only in its reference data. AnthracoForm is an aid to identification, not a substitute for it: reference atlases, comparison collections and time at the microscope remain the basis of every determination.

1. What you need

- Any up-to-date web browser. **Google Chrome or Microsoft Edge, version 86 or later**, are recommended: they let AnthracoForm read and write the CSV file directly in a folder you choose (File System Access API). In other browsers you open the file with **Open CSV** and save it with **Download CSV** (section 3.3).
- The HTML file of a regional module: `AnthracoForm_Saharo-Sindian.html` or `AnthracoForm_Eastern-Mediterranean.html`. Nothing else: no installation, no server, no account and no internet connection.
- A folder for your data, ideally one that is backed up. AnthracoForm writes one CSV file per site in this folder.

1.1 Getting the application

1. Download the latest release from Zenodo (<https://doi.org/10.5281/zenodo.20487673>) or from the Releases page of the GitHub repository (`ignaziominervini-ctrl/AnthracoForm`).
2. Unzip the archive. Each module is in its own folder (Saharo-Sindian/, Eastern-Mediterranean/), with its README, CHANGELOG and LICENSE.
3. Copy the HTML file of your module to your computer and open it in a web browser (double-click, or drag it onto a browser window). Do not open it from inside the zip archive.

The file works offline: it includes everything it needs, including the libraries that produce the PDF record sheet.

2. The screen at a glance

At start-up a welcome window reminds you what the tool is for (Figure 1). Tick *don't show again on startup* to skip it next time.

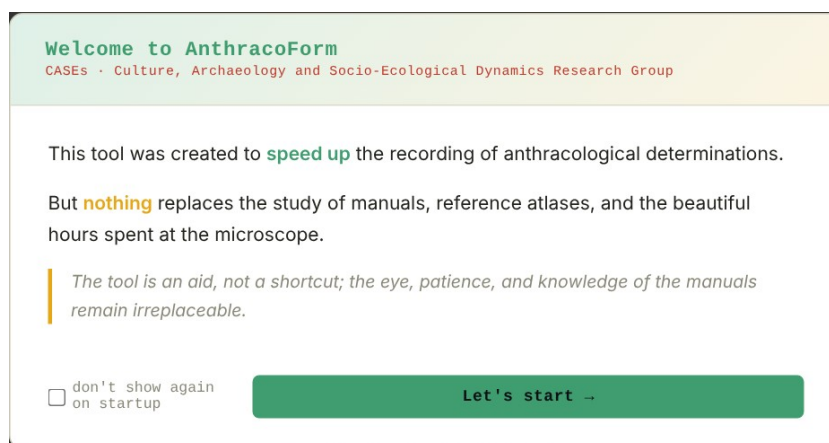


Figure 1. The welcome window.

The screen has three bands (Figure 2):

- **Header:** module name; record navigation (⏮ ⏪ ⏩ ⏭); the position of the current record and the total number of records (*Tot*); ✓ **Save** (or ✓ **Update** when you are editing an existing record) and ✕ **Delete**. The badge on the left says whether you are writing a NEW RECORD or editing one (EDIT n/N).
- **File bar:** the folder and file in use, time of the last save and number of records; **Select folder** (then **Change folder**) and ↻ **Reload CSV**; **FIND**, to jump to a record by SU or by id_det.
- **Tabs:** **Form** (recording), **IAWA Reverse Lookup**, **Dichotomous Key** and **Compare Taxa** (identification tools, section 6).



Figure 2. Header, file bar and tabs, with the demonstration file open.

3. Starting a session

3.1 Opening or creating the site file

1. Click **Select folder** and choose the folder where your data are kept. The browser asks for permission to edit files in that folder: allow it.
2. Type the name of the CSV file for this session, without .csv, and confirm. If a file with that name exists in the folder it is opened; otherwise a new, empty file is created. AnthracoForm proposes the name used last time.
3. The file bar shows the folder and file name and the number of records loaded. The next free id_det is already filled in on the form.

From now on **every save is written to the CSV file on disk immediately**: there is no separate "save file" step, and the file on disk is always up to date. Use one file per site.

3.2 Reloading and changing file

- ↻ **Reload CSV** reads the file again from disk (useful if you edited it elsewhere). Unsaved changes in the form are lost: AnthracoForm asks before reloading.
- **Change folder** opens another folder or file.

3.3 Browsers without folder access

In browsers without the File System Access API the file bar says *No folder access in this browser* and shows an **Open CSV** button. Click it to open an existing file, or start recording straight away for a new one. Records are kept in memory: save them with **Download CSV** (bottom right of the form), which downloads the file with the same name, and replace your copy with the downloaded one. Until you download, the file bar shows *changes not yet downloaded*, and the browser warns you if you try to close the page.

3.4 What the browser remembers

The data are only in the CSV file. The browser stores a few conveniences: the welcome-window setting, the name of the last file, the list of taxa offered under the determination field, and the families and ecological groups you type for taxa that are not in the module's dictionary. These belong to the browser on that computer; clearing the browser data removes them, but never touches your CSV files.

4. Recording a fragment

The Form tab holds one fragment. Fill it in from top to bottom and save. Figure 3 shows the upper part of the form for a saved record.

Figure 3. Sample metadata, wood type, channel, fraction, fracture type and determination.

4.1 Sample metadata

id det. is assigned automatically and gives the order of determination, which the curves follow. Fill in *id samp.* (sample or bag number), *site*, *trench*, *SU*, *layer* and *context*.

KEEP (padlock row): tick a field to keep its value for the next records. Typically you lock site, sample, SU, operator and microscope once per SU and then record only the fragment. The same row lets you keep a **channel** and the **fraction**.

4.2 Wood type, channel, fraction, fracture type

Click a button to select a value; click it again to clear it.

- **Wood type:** *Angiosperm* or *Gymnosperm*. It decides which IAWA list is shown below (hardwood or softwood features).
- **Channel:** where the fragment comes from. The channel decides what the fragment can contribute to (Table 1).
- **Fraction:** >4, 4–2, 2–1 or 1–0.5 mm (stored as gt4, 4to2, 2to1, 1to05).
- **Fracture type:** irregular, cubical, wedge shaped or elongated.

Channel	Meaning	Curves and counts
HF / LF	Heavy and light fraction from flotation of a sample of known volume (typically 20 or 40 L).	Abundances and saturation curve of the SU.

Channel	Meaning	Curves and counts
DS	Dry sieving in the field, fixed mesh, volume not recorded.	List of taxa; conditional discovery curve (CDC).
HC	Hand-collected during excavation, selected by eye.	List of taxa; site-level taxonomic discovery curve (TDC).
STR	Closed context: hearth, oven, dump area, or a charcoal concentration distinct from the surrounding matrix.	List of taxa; analysed by event, never summed with the SU.
C14	Fragments selected for radiocarbon dating.	List of taxa only.

Table 1. Recovery channels. Every determined fragment enters the list of taxa of its SU; only HF and LF enter abundances and the saturation curve.

4.3 Determination

- Type the taxon in **det**. The **TAXA** buttons below list the taxa already recorded in the file: click one to insert it (the small **X** removes it from the list, not from the data).
- **cf.** adds or removes "cf." in the determination. A determination with cf. counts as a fragment but never adds a taxon to the curves; neither do determinations covering several taxa, written with a slash (e.g. *Senegalia/Prosopis* type).
- **undetermined**: tick it for fragments that cannot be identified.
- **family** and **eco** fill in automatically from the taxa dictionary of the module and can be edited. If you type family and eco for a taxon the dictionary does not know, AnthracOForm remembers them for that taxon in this browser.
- A yellow label *new taxon* → *update saturation curve* appears when the taxon is new for the current SU.

4.4 Provisional morphological types

A fragment that you cannot identify yet can be assigned to a **morphological type** (tipo_1 to tipo_20 per file), with a short **type description**. When the type is identified, click **Resolve** → , enter det, family and eco and click **✓ Apply to all**: the determination is written to every fragment of that type. The type field of those fragments is cleared, but the type is not lost: it is kept in the tipo_resolved column, together with its description, and the form shows *resolved from tipo_n*.

4.5 Taphonomy, morphology, observed sections and photo status

The screenshot displays the AnthracOForm interface, which is divided into several sections for data entry. On the left, there are three main panels: 'TAPHONOMY' (TAP), 'MORPHOLOGY' (MOR), and 'OBSERVED SECTIONS' (OBS). The 'TAPHONOMY' panel includes checkboxes for radial crack, wood compression, xylophagous activity, oxidation, mineralisation, cell deformation, and vitrification. The 'MORPHOLOGY' panel includes checkboxes for twig, bark, and curv. The 'OBSERVED SECTIONS' panel includes checkboxes for TRV view, TAN view, RAD view, and PHOTO STATUS. On the right, there are three panels showing the IAWA checklists for transverse (TRV), tangential (TAN), and radial (RAD) sections. Each panel has a 'Filter' button and a list of features with checkboxes. In the TRV panel, features 1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823, 824, 825, 826, 827, 828, 829, 830, 831, 832, 833, 834, 835, 836, 837, 838, 839, 840, 841, 842, 843, 844, 845, 846, 847, 848, 849, 850, 851, 852, 853, 854, 855, 856, 857, 858, 859, 860, 861, 862, 863, 864, 865, 866, 867, 868, 869, 870, 871, 872, 873, 874, 875, 876, 877, 878, 879, 880, 881, 882, 883, 884, 885, 886, 887, 888, 889, 890, 891, 892, 893, 894, 895, 896, 897, 898, 899, 900, 901, 902, 903, 904, 905, 906, 907, 908, 909, 910, 911, 912, 913, 914, 915, 916, 917, 918, 919, 920, 921, 922, 923, 924, 925, 926, 927, 928, 929, 930, 931, 932, 933, 934, 935, 936, 937, 938, 939, 940, 941, 942, 943, 944, 945, 946, 947, 948, 949, 950, 951, 952, 953, 954, 955, 956, 957, 958, 959, 960, 961, 962, 963, 964, 965, 966, 967, 968, 969, 970, 971, 972, 973, 974, 975, 976, 977, 978, 979, 980, 981, 982, 983, 984, 985, 986, 987, 988, 989, 990, 991, 992, 993, 994, 995, 996, 997, 998, 999, 1000. In the TAN panel, features 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823, 824, 825, 826, 827, 828, 829, 830, 831, 832, 833, 834, 835, 836, 837, 838, 839, 840, 841, 842, 843, 844, 845, 846, 847, 848, 849, 850, 851, 852, 853, 854, 855, 856, 857, 858, 859, 860, 861, 862, 863, 864, 865, 866, 867, 868, 869, 870, 871, 872, 873, 874, 875, 876, 877, 878, 879, 880, 881, 882, 883, 884, 885, 886, 887, 888, 889, 890, 891, 892, 893, 894, 895, 896, 897, 898, 899, 900, 901, 902, 903, 904, 905, 906, 907, 908, 909, 910, 911, 912, 913, 914, 915, 916, 917, 918, 919, 920, 921, 922, 923, 924, 925, 926, 927, 928, 929, 930, 931, 932, 933, 934, 935, 936, 937, 938, 939, 940, 941, 942, 943, 944, 945, 946, 947, 948, 949, 950, 951, 952, 953, 954, 955, 956, 957, 958, 959, 960, 961, 962, 963, 964, 965, 966, 967, 968, 969, 970, 971, 972, 973, 974, 975, 976, 977, 978, 979, 980, 981, 982, 983, 984, 985, 986, 987, 988, 989, 990, 991, 992, 993, 994, 995, 996, 997, 998, 999, 1000. In the RAD panel, features 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332,

- **Taphonomy:** tick the alterations present (radial cracks, wood compression, xylophagous activity, oxidation, mineralisation, cell deformation, vitrification). Each is recorded as 1 (present) or 0 (absent).
- **Morphology:** *twig*, *bark*, and *curv* when the ring curvature can be measured to estimate the original diameter.
- **Observed sections:** tick TRV, TAN and RAD for the sections you actually examined. This matters: a feature left unticked counts as absent only where its section was observed.
- **Photo status:** *done* or *to do*.

4.6 IAWA features

The three panels list the IAWA features observable on a charcoal fracture surface, grouped by anatomical category: 102 hardwood features (IAWA Committee 1989) or 77 softwood features (IAWA Committee 2004), depending on the wood type. Tick the features you see. The filter box above each panel searches by number or by text; the number at the top right of each panel counts the features ticked. Features visible in more than one section appear once. The numbers on screen are the IAWA feature numbers; in the CSV each feature is a column with the prefix of its list (e.g. *iawa_A5*, *iawa_G40*).

4.7 Selected codes, photographs, notes and saving

Figure 5. Lower part of the form: list of selected IAWA codes, one photograph per section, notes, operator, microscope, date, PDF record sheet and CSV download.

- **N_iawa** lists the codes ticked for the fragment.
- **Photos:** load one photograph per section. The CSV keeps only the file name; the image stays in your archive. Name your files so that they lead back to the record (for example site, SU, section, id_det and magnification).
- **Notes** (several lines are allowed), **Operator**, **Microscope** and **Date** (today's date by default; stored as YYYY-MM-DD).
- ✓ **Save** (or **Ctrl+S**) writes the record to the CSV file and opens a new empty form with the kept values already in place.

5. Finding and editing records

- Move through the records with **←** **→** or **Alt+←** **Alt+→** (from any tab).
- **FIND**: type an SU and press Enter to jump to its first record; press Enter again to cycle through the records of that SU. Type an id_det and press Enter to jump to that record.
- While a record is open the badge shows EDIT and the save button becomes **✓ Update**. Moving between records never rewrites a record you have not changed.
- **✕ Delete** removes the current record after a confirmation. It cannot be undone.

Keys	Action
Ctrl+S	Save or update the current record
Alt+← / Alt+→	Previous / next record
Enter in FIND	Jump to the SU or id_det typed
Esc	Close the welcome window

Table 2. Keyboard shortcuts.

6. Identification tools

The three tools use the reference data of the module: the taxa with their IAWA profiles and the dichotomous key. They rank and compare candidates; the determination is always entered by the operator.

6.1 IAWA Reverse Lookup

The lookup uses the features ticked on the current fragment (shown at the top; the number on the tab counts them). It lists the taxa of the module ordered by the share of their reference profile that you have observed (*covered*). For each taxon it shows the features that match, the features you ticked that the taxon does not have (*absent in taxon*), and the features of the taxon you have not ticked. Set a minimum number of matching features to hide weak candidates. **✓ Apply to det** copies the taxon into the determination field.

The screenshot displays the IAWA Reverse Lookup tool interface. At the top, 'SELECTED IAWA CODES' are shown as 4, 24, 99, and 102. Below this, a search bar contains 'Lancelotti 2018 - S2' and a 'min' dropdown set to 1. The results are listed in a table-like format with two main entries:

- Tamarix aphylla**: 13% covered, all present. It shows a 'MATCHING' section with codes 4, 24, 99, and 102. Below this, a section 'IN TAXON, NOT CHECKED' lists codes 1 through 137. A green bar at the bottom of this entry contains '✓ Apply to det'.
- Balanites aegyptiaca**: 13% covered, 2 absent. It shows a 'MATCHING' section with codes 99 and 102. Below this, a section 'ABSENT IN TAXON' lists codes 4 and 24. Further down, a section 'IN TAXON, NOT CHECKED' lists codes 2 through 121. A green bar at the bottom of this entry contains '✓ Apply to det'.

At the bottom, a third entry for **Salvadora persica** is partially visible.

Figure 6. IAWA Reverse Lookup for a fragment with four features ticked.

6.2 Dichotomous Key

Follow the key one couplet at a time by clicking the statement that fits the fragment. The path taken is listed on the left; ← **Back** and ↺ **Restart** let you correct it. A result can be copied to the determination field. In the Saharo-Sindian (Sindh) module the key reproduces, couplet by couplet, the published key of Lancelotti (2018, Supplementary Material S2). In the Eastern Mediterranean (Lebanon) module it is the key to genera of Akkemik and Yaman (2012), and each result lists the species of that genus in the reference database; passages of the transcription still to be checked against the original are flagged on screen.

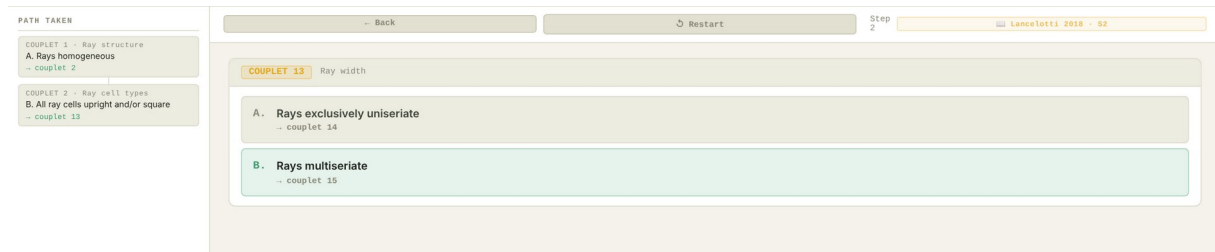


Figure 7. Dichotomous Key after two couplets.

6.3 Compare Taxa

Choose two or three taxa and click **Compare** → . The result lists the key distinguishing characters first, then, section by section, the features found only in one taxon and those shared.

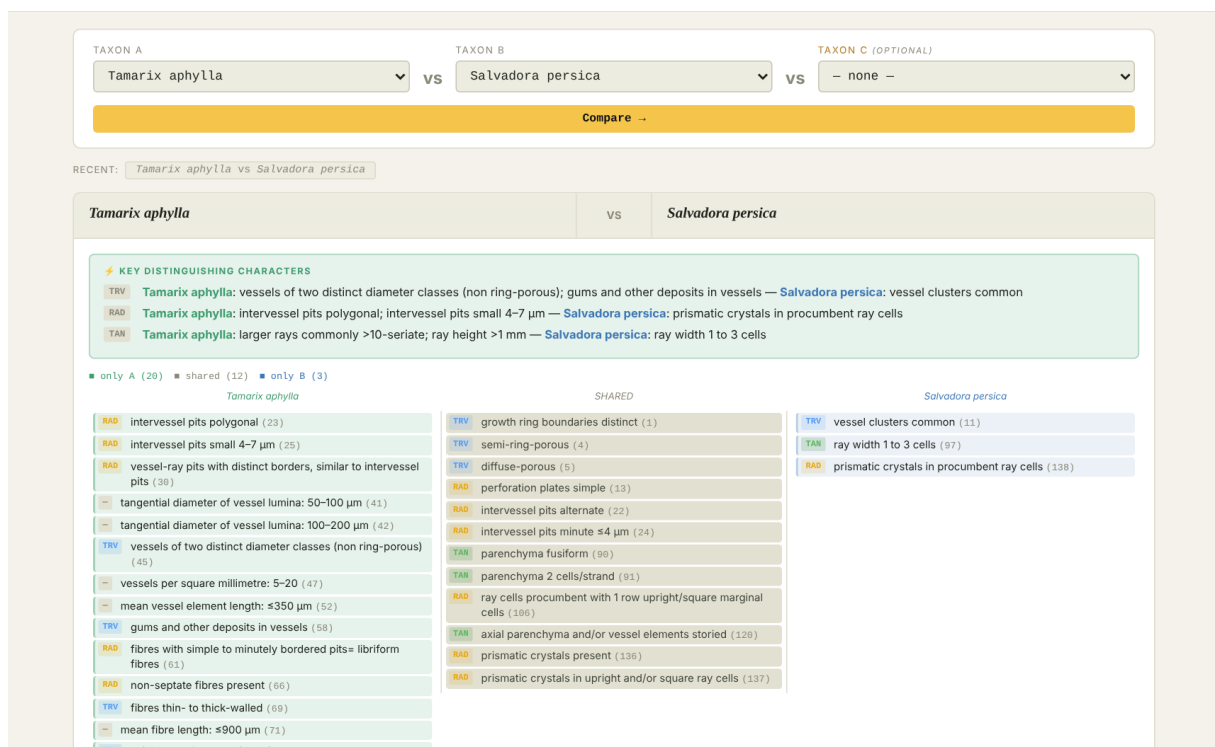


Figure 8. Compare Taxa: two taxa side by side.

7. Checking the sampling during the analysis

7.1 Saturation curve and SU summary

The field *sat. curv.* shows, for HF and LF fragments, how many taxa the SU has reached so far. The  button next to it opens the summary of the current SU, or of the whole assemblage (Figure 9). Its seven tabs are:

Tab	Content
1 · Saturation	Cumulative number of taxa against HF and LF fragments, in order of determination. The bar above the chart shows the current run of fragments without a new taxon and whether the stop criterion (50 fragments without a new taxon) is met.
2 · Richness	HF and LF fragments per taxon. Taxa found in the SU only through other channels (DS, HC, STR, C14) are listed above the chart.
3 · Simpson	Simpson diversity ($1 - D$) of every SU, on HF and LF fragments, for comparison.
4 · Ecology	Composition of HF and LF fragments by the ecological groups defined in the module.
5 · Taphonomy	Frequency of each alteration on the fragments where it was recorded (Figure 10).
6 · Fracture	Distribution of fracture types.
7 · HF/LF	HF and LF fragments per SU.

Table 3. Tabs of the SU summary.

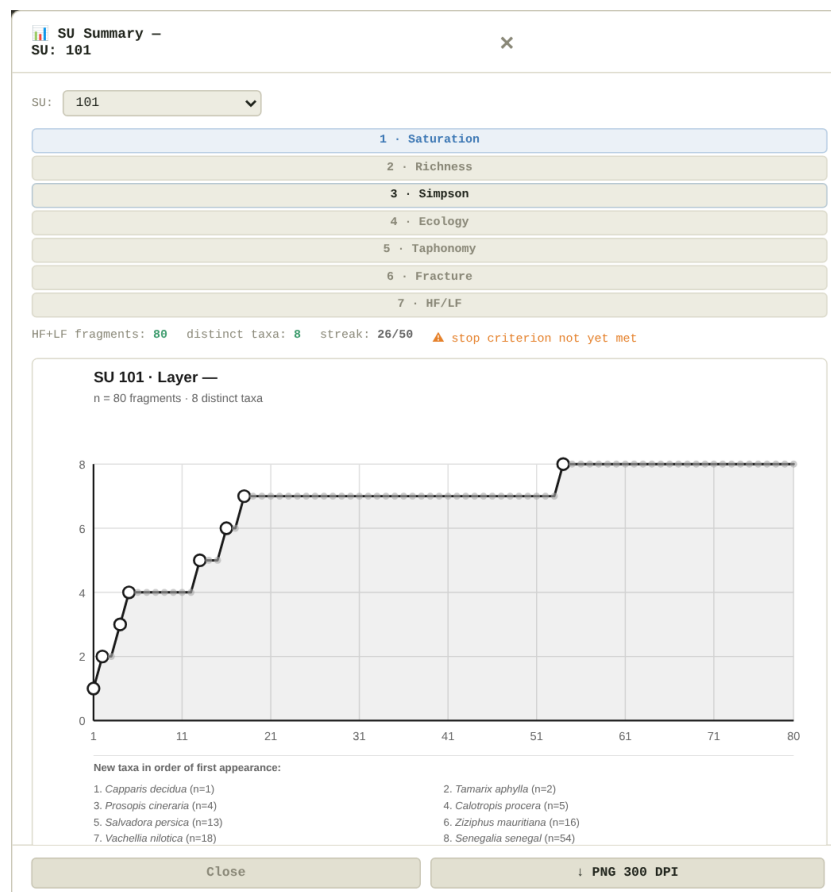


Figure 9. SU summary, Saturation tab. Open circles mark the first fragment of each new taxon; the list under the chart gives the order in which taxa appeared.

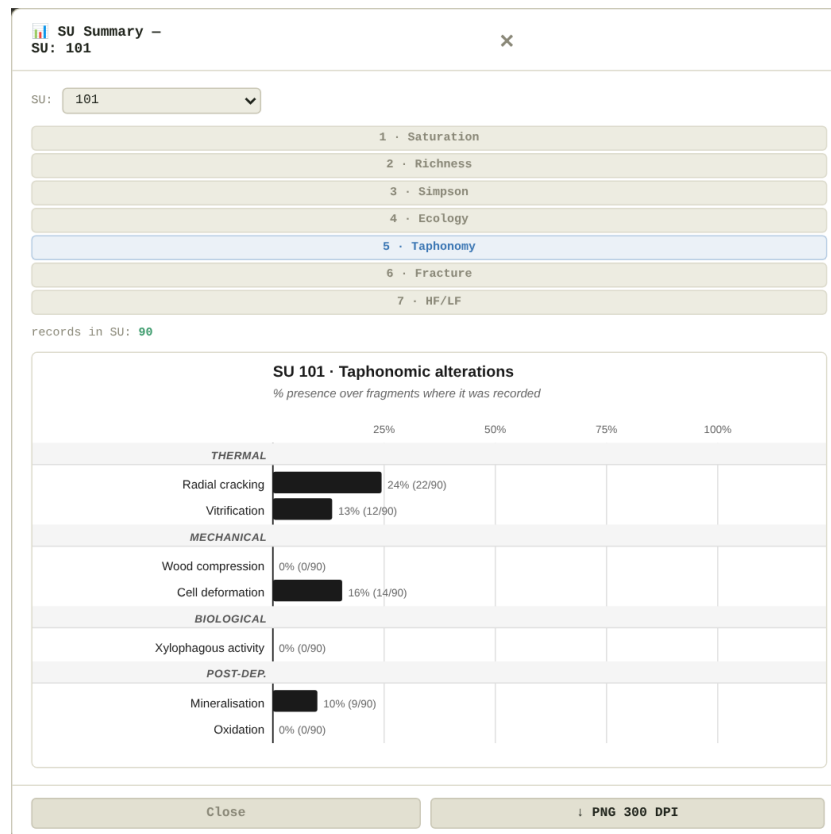


Figure 10. *SU summary, Taphonomy tab.*

Every chart can be exported with **↓ PNG 300 DPI**: black and white, 170 mm wide, ready for print. These summaries are a check during the analysis; analyses for publication are run on the CSV file (S2).

7.2 Curves for the other channels

- **TDC** (taxonomic discovery curve): one curve for the whole site, built on hand-collected (HC) fragments. It records how far the operator has learnt the local woods; it is not a saturation curve.
- **CDC** (conditional discovery curve): for each SU, built on dry-sieved (DS) fragments. It starts from the taxa already found in HF and LF and counts only the taxa that dry sieving adds; a flat curve means that DS adds nothing to flotation, which is itself a result.



Figure 11. Conditional discovery curve (DS) of an SU.

Fragments from closed contexts (STR) and those selected for dating (C14) never enter the curves.

7.3 Picking fragments

The curves are meaningful only if fragments are taken in a way that does not favour one part of the material. We recommend picking, in each SU, the next fragment from the size class and channel (HF or LF, >4 or 4–2 mm) that is least covered so far, and accepting the stop criterion only when each of these cells has at least eight fragments determined. AnthracoForm does not enforce this rule: it belongs to the laboratory protocol.

8. PDF record sheet

 **PDF Report** produces a printable sheet of the current fragment: context, determination, taphonomy, observed sections, the selected features with their IAWA labels, the photographs and the notes. It is meant for doubtful fragments: a second operator can check it, it can be discussed with a specialist or filed with the reference material. It works offline.

9. The data file

Each row of the CSV file is a fragment; the 220 columns are the same in every module. The complete description is in the data dictionary (S1). The essentials:

- UTF-8, comma-separated; 1 = present or yes, 0 = absent or no; **an empty cell means "not recorded"**, never absent.
- Dates in ISO format (YYYY-MM-DD); fractions coded gt4, 4to2, 2to1, 1to05 so that spreadsheets cannot turn them into dates.
- One file per site. Merge the files of a region only when the analyses are finished; site_code + id_det identifies each fragment in the merged data.
- The file can be read directly in R (see S2) or in any statistical program.

Avoid saving the file from Excel. Excel may change the separator to a semicolon, alter dates or drop leading zeros. AnthracoForm reads semicolon-separated files, but the safest course is to open the file in Excel only for reading and to make changes in AnthracoForm.

9.1 Converting existing data

- 1. Start from the empty template supplied with S1 (S1_AnthracoForm_template_v1.3.csv), which has the 220 columns in the right order.
- 2. Copy your data into the matching columns, recoding values to the controlled vocabularies of S1 (channel, fraction, fracture type, 0/1). Leave empty what was not recorded.
- 3. Give each fragment a progressive id_det. If the original order of determination is known, follow it: the curves depend on it.
- 4. Save as CSV (UTF-8), put the file in your data folder and open it with **Select folder**, typing its name. Check a few records on screen.

Files written with earlier versions of AnthracoForm (numeric IAWA columns without prefix, no wood_type, the column ossidation, no tipo_resolved) are converted automatically when opened and rewritten in the current schema at the first save.

10. Regional modules

Every module is the same program. The data that make it regional sit in one block at the top of the HTML file, between the lines REGIONAL MODULE DATA and END OF REGIONAL MODULE DATA. Everything outside that block is identical in all modules, so a correction to the program applies to every region. The block contains:

Element	Content
REGION	Module name and version; default wood type of a new record; the pair shown when Compare Taxa opens; the sources cited on screen and in the citation.
ECO_CATS	The ecological categories of the region, with the colour used in the charts and the rule that assigns an eco value to a category.
TAXA_DICT	Taxa dictionary: for each name, its family and ecological group (used to fill in family and eco).
TAXA_DB	Reference database: each taxon with the list of its IAWA codes (A or G prefix). Used by the Reverse Lookup and Compare Taxa.
KEY_DEF	The dichotomous key: each couplet with its statements, the next couplet or the result.

Table 4. Contents of the regional block.

Module names follow floristic regions (Loidi and Vynokurov 2024); the place in brackets is where the reference collection was made. Until v1.2 the modules were called Sindh and Lebanon.

To build a module for a new region, copy an existing HTML file, replace the content of the block with the data of the region, keeping the same structure, and leave the rest of the file untouched. Then open the file in a browser, walk the whole key and check a few taxa in the Reverse Lookup. The record structure, and therefore the CSV files, stay the same: data from different regions remain comparable.

Palms (Arecaceae) cannot be recorded with the current lists, because their stems have no secondary wood and are identified with features of their own. A third list of features, alongside those for gymnosperms and angiosperms, is in preparation.

11. Troubleshooting

Message or problem	What to do
"No folder access in this browser" in the file bar	The browser cannot write to a folder. Open and save the file with Open CSV and Download CSV (section 3.3), or open the HTML file in Chrome or Edge to save directly in the folder.
The browser asks again for permission to edit the folder	Normal at the start of each session: allow it.
"Data in memory — download CSV at end of session"	The file could not be written (for example because it is open in another program). Close the other program and save again, or download the CSV before closing the page.
"CSV with ";" detected"	The file was saved by a spreadsheet with European settings. It is read correctly and rewritten with commas at the next save.
The welcome window no longer appears	It was switched off with "don't show again on startup". Clearing the browser data brings it back.

Table 5. Common messages and problems.

References

- Akkemik, Ü and Yaman, B. 2012. *Wood Anatomy of Eastern Mediterranean Species*. Remagen-Oberwinter: Kessel.
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- IAWA Committee. 2004. IAWA list of microscopic features for softwood identification. Richter, HG, Grosser, D, Heinz, I and Gasson, PE (eds.). *IAWA Journal*, 25: 1–70.
- Lancelotti, C. 2018. 'Not all that burns is wood'. A social perspective on fuel exploitation and use during the Indus urban period (2600–1900 BC). *PLoS ONE*, 13: e0192364. DOI: <https://doi.org/10.1371/journal.pone.0192364>
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